

DATA CLEANING & REPORTING AUTOMATION

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1. Introduction

This project focuses on Data Cleaning and Reporting Automation using Python. The project demonstrates how to clean missing values, remove duplicate records, generate reports, and visualize student performance data using Pandas and Matplotlib.

2. Objectives

Load and analyze CSV datasets
Handle missing values automatically
Remove duplicate records
Generate statistical summaries
Create visual reports using graphs
Save cleaned datasets automatically

3. Dataset Used

ID	Name	Age	Marks	City
1	Aman	20	85	Bhubaneswar
2	Riya	21	90	Cuttack
3	Rohit		78	Puri
4	Sneha	22	88	Bhubaneswar
5	Aman	20	85	Bhubaneswar
6	Priya	21		Cuttack

4. Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("student_data.csv")

data.columns = data.columns.str.strip()

data['Age'] = pd.to_numeric(data['Age'], errors='coerce')
data['Marks'] = pd.to_numeric(data['Marks'], errors='coerce')

data['Age'] = data['Age'].fillna(data['Age'].mean())
data['Marks'] = data['Marks'].fillna(data['Marks'].mean())

data['City'] = data['City'].fillna("Unknown")

data = data.drop_duplicates()

print(data.describe())

plt.bar(data['Name'], data['Marks'])
plt.show()
```

5. Working Procedure

The program first reads the CSV dataset using Pandas. Then it checks missing values and replaces them using mean values. Duplicate rows are removed to improve data quality. After cleaning, a summary report is generated and a bar chart is created to visualize student marks.

6. Output

Missing values handled successfully Duplicate rows removed Cleaned dataset saved automatically Summary statistics generated Bar chart visualization displayed

7. Conclusion

This project successfully demonstrates the process of Data Cleaning and Reporting Automation using Python. It improves dataset quality, automates reporting tasks, and helps in visual analysis of student performance data.